

characterized to permit application of mathematical modeling techniques. As a result, the fire suppression aspects of sprinkler system design are empirical at best.

Some, but not all, of the current sprinkler system design criteria are based on full-scale testing, including the criteria originally developed for NFPA 231C, NFPA 13D, and parts of NFPA 13, such as the provisions pertaining to the use of CMSA (control mode specific application) and ESFR (early suppression fast response) sprinklers. Most of the protection criteria of NFPA 13 and other sprinkler system design standards, however, are the result of the evolution and application of experienced judgment and intuitive reasoning. In the 1970s, the capabilities of pipe schedule systems, which had demonstrated a hundred years of satisfactory performance, were codified into a system of density/area curves [3]. This permitted the introduction of hydraulic calculations to what had become a cookbook-type method of designing sprinkler systems. Hydraulic calculations allowed system designers to take advantage of strong water supplies to produce more economical systems. It also permitted the determination of specific flows and pressures available at various points of the system, opening the door to the development and use of new types of sprinklers. The number of different types of sprinklers available in the global marketplace has increased dramatically in the past few decades, and the fire protection engineer needs to be knowledgeable with regard to the variety of fire sprinkler products.

Even where design criteria were developed on the basis of full-scale testing, the number and sequence of operating sprinklers has been found to be variable. Some of this variability is due to the variability in fire growth itself and prevailing air currents within a building, but there are also observed phenomena such as “sprinkler skipping,” in which a sprinkler will operate substantially prior to a nearby sprinkler that is closer to the fire plume. Generally associated with high challenge fire scenarios, skipping has been attributed to water drop impingement from nearby operating sprinklers [4], and there is evidence that steps can be taken to shield a

sprinkler’s heat sensing element from such impingement [5].

Because of this history, the calculation methods available to the fire protection engineer in standard sprinkler system design are only ancillary to the true function of a sprinkler system. The sections that follow in this chapter address hydraulic calculations of flow through piping, simple calculations commonly performed in determining water supply requirements, and optional calculations that may be performed with regard to hanging and bracing of system piping. The final section of this chapter deals with the performance of a system relative to a fire, and the material contained therein is totally outside the realm of standard practice. This material is not sufficiently complete to permit a full design approach, but only isolated bits of total system performance. The technical reference guide to the current version of the Fire Dynamics Simulator (FDS), a computational fluid dynamics model that has become the most popular tool in fire protection engineering, includes the statement that “simulating the effects of a sprinkler spray involves a number of elements beyond just activation: computing the droplet trajectories and tracking the water as it drips onto the burning surface” [6]. Efforts are underway to develop a mathematical model of sprinkler spray discharge based on first principles [7], with the hope of ultimately being able to model the impact of a fire sprinkler system on a fire.

Hydraulic Calculations

Density-Based Sprinkler Demand

Occupancy hazard classification, or commodity classification in the case of protecting storage, is the most critical aspect of the sprinkler system design process. If the hazard or commodity class is underestimated, it is possible for fire to overpower the sprinklers, conceivably resulting in a large loss of property or life. Hazard classification is not an area in which calculation methods are presently in use, however. The proper classification of hazard requires experienced judgment

and familiarity with relevant design and installation standards, and an understanding of sprinkler system performance. Commodity classification is also not possible by means of calculation, but intermediate scale fire testing is sometimes used as the basis of classifying a specific product.

Once the hazard or commodity classification is determined and a sprinkler spacing and piping layout has been proposed in conformance with the requirements of the applicable standard, the system designer can initiate a series of calculations to demonstrate that the delivery of a prescribed rate of water application will be accomplished for the maximum number of sprinklers that might be reasonably expected to operate. This number of sprinklers, which must be supplied regardless of the location of the fire within the building, is the basis of the concept of the remote design area. The designer needs to demonstrate that the shape and location of the sprinkler arrangement in the design area will be adequately supplied with water in the event of a fire. Adequacy of water relates to flow, pressure and quantity of water available through the sprinklers expected to operate in response to the fire.

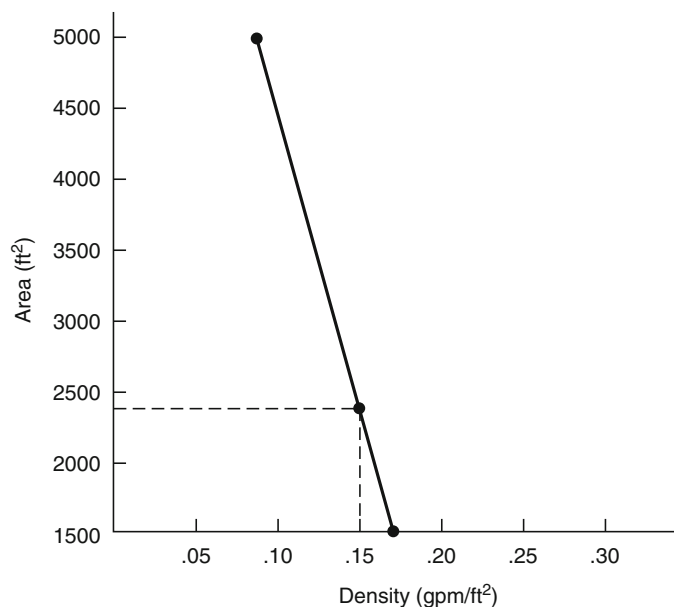
Prior to locating the design area, there is the question of how many sprinklers are to be

included in the expected maximum simultaneous flow area. This question is primarily addressed by the occupancy hazard classification, but the designer also has some freedom to decide this matter.

For example, in the 2013 edition of NFPA 13, Figure 11.2.3.1.1 contains density/area curves from which the designer can select a design area and density appropriate for the occupancy hazard classification. Any point on or to the right of the curve in the figure is acceptable. The designer may select a high density over a small area, or a low density over a large area. In either event, the fire is expected to be controlled by the sprinklers within that design area, without opening any additional sprinklers. For the protection of storage, similar curves or individual density/area specifications can be found in Chaps. 12, 13, 14, 15, 16, 17, 18, 19, 20, and 21. Chapter 22 of NFPA 13 (2013) contains some sprinkler design density/area specifications or hazard classifications from other NFPA codes and standards relating to the protection of specific occupancies.

Example 1 Using the sample density/area curve shown in Fig. 42.1, many different design criteria could be selected, ranging from a density of

Fig. 42.1 Sample density/area curve



0.1 gpm/ft² (3.7 mm/min) over 5000 ft² (500 m²) to 0.17 gpm/ft² (6.9 mm/min) over 1500 ft² (139 m²). Either of these two points, or any point to the right of the curve (such as 0.16 gpm/ft² [6.5 mm/min] over 3000 ft² [276 m²]) would be considered acceptable. A selection of 0.15 gpm/ft² (6.1 mm/min) over 2400 ft² (221 m²) is indicated.

Water is provided only for the number of sprinklers in the design area, since no water is needed for the sprinklers that are not expected to open. The design area will be located in the hydraulically most demanding portion of the system. The actual number of sprinklers in the design area depends, of course, on the spacing of the individual sprinklers. NFPA 13 requires that the design area be divided by the maximum sprinkler spacing used, and that any fractional result be rounded up to the next whole sprinkler, as illustrated below.

Example 2 Based on the point selected from the sample density/area curve above and the proposed maximum spacing of sprinklers, the number of sprinklers to be included in the design area can be determined. If sprinklers are spaced at 12 × 15 ft (3.66 × 4.57 m) so as to each protect an area of 180 ft² (16.72 m²), the design area of 2400 ft² (221 m²) would include.

$$\frac{2,400}{180} = 13.33 = 14 \text{ sprinklers}$$

The remote design area is required to have a rectangular shape, with the long side of the rectangle parallel to the branch lines. The length of the long side of the design area needs to be determined to calculate the number of sprinklers to be included on each branch line in the design area. This length is normally determined by multiplying the square root of the design area by a factor of 1.2 in accordance with the rules of NFPA 13. Again, any fractional result is rounded to the next whole sprinkler as indicated below. Note that other standards or design methods may use multiplication factors other than 1.2. The higher the factor, the more conservative the design, since it requires more

water to be available through individual branch lines.

Example 3 If the 14 sprinklers from Example 2 were spaced 12 ft (3.66 m) along the branch lines and the branch lines were spaced 15 ft (4.57 m) apart, the number of sprinklers along the length of the branch lines in the design area would be

$$\frac{1.2(2,400)^{1/2}}{12} = \frac{1.2(49)}{12} = 4.9 = 5 \text{ sprinklers}$$

If the sprinklers were spaced 15 ft (4.57 m) along the branch lines with the branch lines spaced 12 ft (3.66 m) apart, the design area rectangle would include only 4 sprinklers along its length.

NFPA 13 (2013) contains some exceptions to this method of locating a remote design area and determining the number of sprinklers to be supplied. Chapters 11 and 12 of the standard include special adjustments to the design area based on factors such as the use of a dry system, the use of quick response sprinklers under flat smooth ceilings of limited height, and the existence of nonsprinklered combustible concealed spaces within the building. These chapters also contain rules for the use of a room design method, which can reduce the number of sprinklers expected to operate in a highly compartmented occupancy. Also, beginning in 1985, the standard adopted a four sprinkler design area for dwelling units and their adjacent corridors when residential sprinklers are installed in accordance with their listing requirements. Listing requirements are specific to the applicable standard but normally pertain to the independent laboratory evaluation of performance of a product, system or service, in this case a residential sprinkler.

A step-by-step hydraulic calculation procedure is usually applied. For example, see Chapter 23 of the 2013 edition of NFPA 13. The starting point is the most remote sprinkler in the design area. For tree systems, in which each sprinkler is supplied from only one

direction, the most remote sprinkler is generally the end sprinkler on the farthest branch line from the system riser. This sprinkler, and all others in the design area, must be provided with a sufficient flow of water to meet the density appropriate for the point selected on the density/area curve.

Where a sprinkler protects an irregular area, NFPA 13 prescribes that the area of coverage for the sprinkler must be based on the largest sides of its coverage. In other words, the area which a sprinkler protects for calculation purposes is equal to

$$\text{Area of coverage} = S \times L$$

where S is twice the larger of the distances to the next sprinkler (or wall for an end sprinkler) in both the upstream and downstream directions, and L is twice the larger of the distances to adjacent branch lines (or wall in the case of the last branch line) on either side. This reflects the need to flow more water with increasing distance from the sprinkler, since increased flow tends to expand the effective spray umbrella of the sprinkler.

The minimum flow from a sprinkler must be the product of the area of coverage multiplied by the minimum required density

$$Q = \text{Area of coverage} \times \text{Density}$$

Most of the special listed sprinklers and residential sprinklers have a minimum flow requirement associated with their listings, which is often based on the spacing at which they are used. These minimum flow considerations override the minimum flow based on the density/area method.

Example 4 If a standard spray sprinkler protects an area extending to 7 ft (2.1 m) on the north side (half the distance to the next branch line), 5 ft (1.5 m) on the south side (to a wall), 6 ft (1.8 m) on the west side (half the distance to the next sprinkler on the branch line), and 4 ft (1.2 m) on the east side (to a wall), the minimum flow required for the sprinkler to achieve the density requirement selected in Example 1 can be found

by completing two steps. The first step involves determining the area of coverage. In this case:

$$S \times L = 2(6 \text{ ft}) \times 2(7 \text{ ft}) = 12 \text{ ft} \times 14 \text{ ft} \\ = 168 \text{ ft}^2 (15.6 \text{ m}^2)$$

The second step involves multiplying this coverage area by the required density:

$$Q = A \times \rho = 168 \text{ ft}^2 \times 0.15 \text{ gpm/ft}^2 \\ = 25.2 \text{ gpm} (95.4 \text{ L/min})$$

Pressure Requirements of the Most Remote Sprinkler

When flow through a sprinkler orifice takes place, the energy of the water changes from the potential energy of pressure to the kinetic energy of flow. A formula can be derived from the basic energy equations to determine how much water will flow through an orifice based on the water pressure inside the piping at the orifice:

$$Q = 29.83 c_d d^2 P^{1/2}$$

This formula contains a factor, c_d , which is a discharge coefficient characteristic of the sprinkler orifice that is determined by laboratory testing. For sprinklers, the product listing organizations determine the orifice discharge coefficient for each particular model of sprinkler. To simplify matters, all physical factors of the sprinkler orifice are lumped into what is experimentally determined as the K-factor of a sprinkler, such that

$$Q = K \times P^{1/2}$$

where K has units of $\text{gpm}/(\text{psi})^{1/2}$ [$\text{L/min}/(\text{bar})^{1/2}$].

Since the required minimum flow at the most remote sprinkler is known, determined by either the density/area method or the special sprinkler listing, the minimum pressure needed at the most remote sprinkler can easily be calculated.

Since $Q = K(P)^{1/2}$ then $P = (Q/K)^2$

Many sprinklers require a minimum operating pressure of 7 psi (0.48 bar) to ensure proper spray pattern development.

Example 5 The pressure required at the sprinkler in Example 4 can be determined using the above formula once a specific sprinkler is selected. The K-factor for any sprinkler needs to be confirmed with the manufacturer and listing organization. For the purposes of this example, a K-factor of 5.6 is assumed, since this is a common value for what are considered standard ½-inch orifice sprinklers.

$$P = \left(\frac{Q}{K}\right)^2 = \left(\frac{25.2}{5.6}\right)^2 = 20.2 \text{ psi}(1.4 \text{ bar})$$

Once the minimum pressure at the most remote sprinkler is determined, the hydraulic calculation method proceeds backward toward the source of water supply. If the sprinkler spacing is regular, it can be assumed that all other sprinklers within the design area will be flowing at least as much water, and the minimum density is assured. If sprinkler spacing is irregular due to walls and obstruction, or sprinklers with different K-factors are used, it must be verified that each sprinkler is provided with sufficient flow.

As the calculations proceed toward the system riser and water supply, the minimum pressure requirements increase, because additional pressures are needed to overcome losses associated with elevation changes, pipe friction, and turbulence caused by fittings, so that the minimum design densities for all sprinklers in the design area are maintained. The determination of the friction and elevation losses are discussed below, with their values added to the total pressure requirements.

It should be noted that each sprinkler closer to the source of supply will show a successively greater flow rate, since a higher total pressure is available at that point in the system piping. This effect on the total water demand is termed hydraulic increase, and is the reason why the total water demand of a system is not simply equal to the product of the minimum density and the design area. When calculations are complete, the sprinkler system demand will be known, stated in the form of a specific flow at a specific pressure. A hose demand is also sometimes added to the sprinkler system demand.

The total quantity of water will be determined based upon the duration of required flow that will be specified by applicable standards.

Pressure Losses Through Piping, Fittings, and Valves

Friction losses resulting from water flow through piping can be estimated by several engineering approaches, but the most common is the Hazen-Williams method. This approach is based on the formula developed empirically by Hazen and Williams:

$$p = \frac{4.52Q^{1.85}}{C^{1.85}d^{4.87}}$$

- where
- p* = Friction loss per ft of pipe in psi
- Q* = Flow rate in gpm
- d* = Internal pipe diameter in in.
- C* = Hazen-Williams coefficient

The choice of *C* is critical to the accuracy of the friction loss determination, and is therefore normally stipulated by design standards. *C* values for various types of pipe materials are shown in Fig. 42.1. The values assigned for use are intended to simulate the expected interior roughness of aged pipe (Table 42.1).

Rather than make the Hazen-Williams calculation for each section of piping, it has become standard practice, when doing hand calculations, to use a friction loss table, which contains all values of *p* for various values of *Q* and various pipe sizes. In many cases the tables are based on

Table 42.1 C values for pipes

Type of pipe	Assigned C factor
Steel pipe—dry and preaction systems	100
Steel pipe—wet and deluge systems	120
Galvanized steel pipe—dry and preaction systems	100
Galvanized steel pipe—wet and deluge systems	120
Cement lined cast or ductile iron	140
Copper tube	150
Plastic (listed)	150

the use of Schedule 40 steel pipe for wet systems. The use of other pipe schedules, pipe materials, or system types may require the use of multiplying factors. Most commercially available sprinkler system hydraulic calculation programs have these values programmed into the software.

Once the value of friction loss per foot is determined using either the previous equation or friction loss tables, the total friction loss through a section of pipe is found by multiplying p by the length of pipe, L . Since some standards use p to designate loss per foot, total friction loss in a length of pipe can be designated by p_f , where

$$p_f = p \times L$$

In the analysis of complex piping arrangements, it is sometimes convenient to lump the values of all factors in the Hazen-Williams equation (except flow) for a given length of pipe into a constant, K , identified as a friction loss coefficient. To avoid confusion with the nozzle coefficient K , this coefficient can be identified as FLC, friction loss coefficient.

$$\text{FLC} = \frac{(L \times 4.52)}{(C^{1.85} d^{4.87})}$$

The value of p_f is therefore equal to

$$p_f = \text{FLC} \times Q^{1.85}$$

Example 6 If the most remote sprinkler on a branch line requires a minimum flow of 25.2 gpm (92.1 L/min) for a minimum pressure of 20.2 psi (1.4 bar) as shown in Examples 4 and 5, and the second sprinkler on the line is connected by a 12 ft (3.6 m) length of 1 in. (25.4 mm) Schedule 40 steel pipe, with both sprinklers mounted directly in fittings on the pipe (no drops or sprigs), the minimum pressure required at the second sprinkler can be found by determining the friction loss caused by a flow of 25.2 gpm (92.1 L/min) through the piping to the end sprinkler. Typically, pressure losses associated with straight-through fittings can be ignored in the calculation process if there is no change in flow direction. Also, the fitting directly attached to each sprinkler is generally ignored, since K -factors of sprinklers are determined with such a fitting in place.

Using the Hazen-Williams equation with values of 25.2 for Q , 120 for C , and 1.049 for d (the inside diameter of Schedule 40 steel 1 in. pipe) results in a value of $p = 0.20$ psi (0.012 bar) per foot of pipe. Multiplying by the 12 ft (3.6 m) length results in a total friction loss of $p_f = 2.4$ psi (0.17 bar). The total pressure required at the second sprinkler on the line is therefore $20.2 \text{ psi} + 2.4 \text{ psi} = 22.6 \text{ psi}$ (1.6 bar). This will result in a flow from the second sprinkler of $Q = K(P)^{1/2} = 26.6 \text{ gpm}$ (100.7 L/min).

Minor losses through fittings and valves are not friction losses but energy losses, caused by turbulence in the water flow, which increase as the velocity of flow increases. Nevertheless, it has become standard practice to simplify calculation of such losses through the use of “equivalent lengths,” which are added to the actual pipe length in determining the pipe friction loss. NFPA 13 contains a table of equivalent pipe lengths for this purpose (Table 42.2). As an example, if a 2 in. (50.8 mm) 90-degree long turn elbow is assigned an equivalent length of 3 ft (0.914 m), this means that the energy loss associated with turbulence through the elbow is expected to approximate the energy loss to friction through 3 ft of 2 in. pipe (0.914 m of 50.8 mm pipe). As with the friction loss tables, the equivalent pipe length chart is based on the use of steel pipe with a C -factor of 120, and the use of other piping materials requires multiplying factors. The equivalent pipe length for pipes having C values other than 120 should be adjusted using the following multiplication factors: 0.713 for a C value of 100; 1.16 for a C value of 130; 1.33 for a C value of 140; 1.51 for a C value of 150.

Example 7 If the 12 ft (3.6 m) length of 1 in. (25.4 mm) pipe in Example 6 had contained four elbows so as to avoid a building column, the pressure loss from those elbows could be approximated by adding an equivalent length of pipe to the friction loss calculation. Table 42.2 gives a value of 2 ft (0.610 m) as the appropriate equivalent length for standard elbows in 1 in. (25.4 mm) Schedule 40 steel pipe. For four elbows, the equivalent fitting length would be 8 ft (2.4 m). Added to the actual pipe length of

Table 42.2 Equivalent pipe length chart (for $C = 120$)

Fittings and valves	Fittings and valves expressed in equivalent feet of pipe													
	¾ in.	1 in.	1¼ in.	1½ in.	2 in.	2½ in.	3 in.	3½ in.	4 in.	5 in.	6 in.	8 in.	10 in.	12 in.
45° elbow	1	1	1	2	2	3	3	3	4	5	7	9	11	13
90° standard elbow	2	2	3	4	5	6	7	8	10	12	14	18	22	27
90° long turn elbow	1	2	2	2	3	4	5	5	6	8	9	13	16	18
Tee or cross (flow turned 90°)	3	5	6	8	10	12	15	17	20	25	30	35	50	60
Butterfly valve	—	—	—	—	6	7	10	—	12	9	10	12	19	21
Gate valve	—	—	—	—	1	1	1	1	2	2	3	4	5	6
Swing check ^a	—	5	7	9	11	14	16	19	22	27	32	45	55	65

For SI units: 1 ft = 0.3048 m

^aDue to the variations in design of swing check valves, the pipe equivalents indicated in the above chart are to be considered average

12 ft (3.6 m), the total equivalent length would be 20 ft (6 m). This results in a new value of $p_f = 20 \text{ ft} \times 0.20 \text{ psi/ft} = 4.0 \text{ psi}$ (0.28 bar). The total pressure at the second sprinkler would then be equal to $20.2 \text{ psi} + 4.0 \text{ psi} = 24.2 \text{ psi}$ (1.67 bar). The total flow from the second sprinkler in this case would be $Q = K(P)^{1/2} = 27.5 \text{ gpm}$ (100.4 L/min).

Some types of standard valves, such as swing check valves, are included in Table 42.2, the equivalent pipe length chart. Equivalent lengths for pressure losses through system alarm, dry, and deluge valves are determined by the approval laboratories at the time of product listing.

Use of Velocity Pressures

The value of pressure, P , in the sprinkler orifice flow formula can be considered either the total pressure, P_t , or the normal pressure, P_n , since design standards typically permit the use of velocity pressures at the discretion of the designer. Total pressure, normal pressure, and velocity pressure, P_v , have the following relationship:

$$P_n = P_t - P_v$$

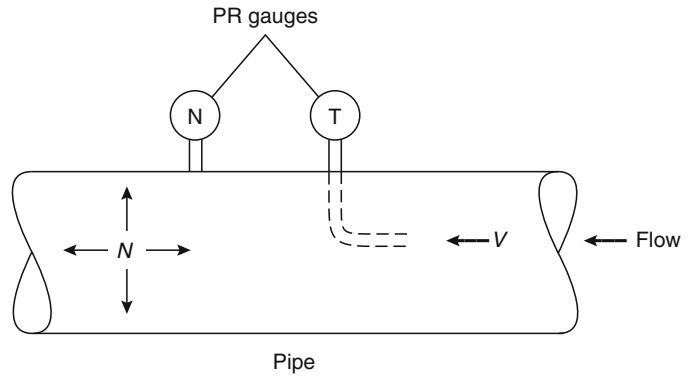
Total pressure is the counterpart of total energy or total head, and can be considered the pressure that would act against an orifice if all of the energy of the water in the pipe at that point were focused toward flow out of the orifice. This

is the case where there is no flow past the orifice in the piping. Where flow does take place in the piping past an orifice, however, normal pressure is the portion of the total pressure acting perpendicular to the direction of flow in the piping, and therefore acting in the direction of flow through the orifice. The amount by which normal pressure is less than total pressure is velocity pressure, which is acting in the direction of flow in the piping. Velocity pressure corresponds to velocity energy, which is the energy of motion. There is no factor in the above expression for elevation head, because the flow from an orifice can be considered to take place in a datum plane.

When velocity pressures are used in calculations, it is recognized that some of the energy of the water is in the form of velocity head, which is not acting normal to the pipe walls (where it would help push water out the orifice), but rather in the downstream direction. Thus, for every sprinkler (except the end sprinkler on a line), slightly less flow takes place than what would be calculated from the use of the formula $Q = K(P)^{1/2}$ (Fig. 42.2).

Design standards typically permit the velocity pressure effects to be ignored, however, since they are usually rather minor for most sprinkler system configurations. Additionally, ignoring the effects of velocity pressure tends to produce a more conservative design in that the calculated system demand (flow and pressure) increase when velocity pressures are not taken into account.

Fig. 42.2 Velocity and normal pressures in piping



If velocity pressures are considered, normal pressure rather than total pressure is used when determining flow through any sprinkler except the end sprinkler on a branch line, and through any branch line except the end branch line on a cross main. The velocity pressure, P_v , which is subtracted from the total pressure in order to determine the normal pressure, is determined as

$$P_v = \frac{v^2}{2g} \times 0.433 \text{ psi/ft (0.098 bar/m)}$$

or

$$P_v = 0.001123Q^2/d^4$$

where Q is the upstream flow through the piping to an orifice (or branch line) in gpm and d is the actual internal diameter of the upstream pipe in inches.

Because design standards typically mandate the use of the upstream flow, an iterative approach to determining the velocity pressure is necessary. The upstream flow cannot be determined unless the flow from the sprinkler (or branch line) in question is known. Since the flow from the sprinkler (or branch line) is affected by the velocity pressure resulting from the upstream flow, an estimate of the upstream flow is needed to start the iteration.

Example 8 If the pipe on the upstream side of the second sprinkler in Example 6 were 3 in. Schedule 40 steel pipe with an inside diameter of 1.38 in. (35 mm), the flow from the second sprinkler would be considered to be 26.6 gpm (100.2 L/min) as determined at the end of

Example 6, if velocity pressures were not included.

If velocity pressures were to be considered, an upstream flow would first be assumed. Since the end sprinkler had a minimum flow of 25.2 gpm (95.2 L/min) and the upstream flow would consist of the combined flow rates of the two sprinklers, an estimate of 52 gpm (196.8 L/min) appears reasonable. Substituting this flow and the pipe diameter into the equation for velocity pressure gives

$$\begin{aligned} P_v &= \frac{0.001123Q^2}{d^4} \\ &= \frac{0.001123(52)^2}{(1.38)^4} \\ &= 0.8 \text{ psi (0.06 bar)} \end{aligned}$$

This means that the actual pressure acting on the orifice of the second sprinkler is equal to

$$\begin{aligned} P_n &= P_t - P_v \\ &= 22.6 \text{ psi} - 0.8 \text{ psi} \\ &= 21.8 \text{ psi (1.50 bar)} \end{aligned}$$

This would result in a flow from the second sprinkler of

$$\begin{aligned} Q &= K(P)^{1/2} \\ &= 26.1 \text{ gpm (98.7 L/min)} \end{aligned}$$

Combining this flow with the known flow from the end sprinkler results in a total upstream flow of 51.3 gpm (194.2 L/min). To determine if the initial guess was close enough, determine the velocity pressure that would result from an upstream flow of 51.3 gpm (194.2 L/min).

This calculation also results in a velocity pressure of 0.8 psi (0.06 bar), and the process is therefore complete. It can be seen that the second sprinkler apparently flows 0.5 gpm (1.9 L/min) less than the estimated flow due to velocity pressures.

Elevation Losses

Variation of pressure within a fluid at rest is related to the density or unit (specific) weight of the fluid. The unit weight of a fluid is equal to its density multiplied by the acceleration of gravity. The unit weight of water is 62.4 lbs/ft³ (1000 kg/m³).

This means that one cubic foot of water at rest weighs 62.4 lbs (1000 kg). The cubic foot of water, or any other water column one foot high, thus results in a static pressure at its base of 62.4 lbs/ft² (304.66 kg/m²). Divided by 144 in.² per ft² (1.020×10^4 kg/m² bar), this results in a pressure of 0.433 lb per in.² per ft (0.099 bar/m) of water column.

A column of water 10 ft (3.048 m) high similarly exerts a pressure of $10 \text{ ft} \times 62.4 \text{ lbs/ft}^2 \times 1 \text{ ft}^2/144 \text{ in.}^2 = 4.33 \text{ psi}$ ($3.048 \text{ m} \times 999.5 \text{ kg/m}^2 \div 1.020 \times 10^4 \text{ kg/m}^2 \text{ bar} = 0.299 \text{ bar}$). The static pressure at the top of both columns of water is equal to zero (gauge pressure), or atmospheric pressure.

On this basis, additional pressure must be available within a sprinkler system water supply to overcome the pressure loss associated with elevation, i.e. when water flow is acting against the force of gravity. This pressure is equal to 0.433 psi/ft (0.099 bar/m) of elevation of the sprinklers above the level where the water supply information is known.

Sometimes the additional pressure needed to overcome elevation is added at the point where the elevation change takes place within the system. If significant elevation changes take place within a portion of the system that is likely to be considered as a representative flowing orifice (such as a single branch line along a cross main that is equivalent to other lines in the remote design area), then it is considered more accurate to wait until calculations have been completed, and simply add an elevation pressure increase to

account for the total height of the highest sprinklers above the supply point.

Example 9 The pressure that must be added to a system supply to compensate for the fact that the sprinklers are located 120 ft (36.6 m) above the supply can be found by multiplying the total elevation difference by 0.433 psi/ft (0.099 bar/m).

$$120 \text{ ft} \times 0.433 \text{ psi/ft} = 52 \text{ psi} \text{ (3.62 bar)}$$

Loops and Grids

Hydraulic calculations become more complicated when piping is configured in loops or grids, such that water feeding any given sprinkler or branch line can be supplied through more than one route. A number of computer programs that can quickly complete the repetitive calculations have therefore been developed specifically for fire protection systems, and are being marketed commercially.

Determining the flow split that takes place in the various parts of any loop or grid is accomplished by applying the basic principles of conservation of mass and conservation of energy. For a single loop, it should be recognized that the energy loss across each of the two legs from one end of the system to the other must be equal. Otherwise, a circulation would take place within the loop itself. Also, mass is conserved by the fact that the sum of the two individual flows through the paths is equal to the total flow into (and out of) the loop (Fig. 42.3).

Applying the Hazen-Williams formula to each leg of the loop

$$p_f = L_1 \frac{4.52 Q_1^{1.85}}{C_1^{1.85} d_1^{4.87}} = L_2 \frac{4.52 Q_2^{1.85}}{C_2^{1.85} d_2^{4.87}}$$

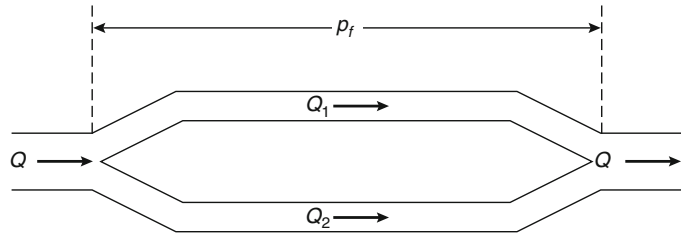
Substituting the term FLC for all terms except Q ,

$$p_f = \text{FLC}_1 Q_1^{1.85} = \text{FLC}_2 Q_2^{1.85}$$

This simplifies to become

$$\left(\frac{Q_1}{Q_2} \right)^{1.85} = \frac{\text{FLC}_2}{\text{FLC}_1}$$

Fig. 42.3 Example of a simple loop configuration



Since Q_1 and Q_2 combine to create a total flow of Q , the flow through one leg can be determined as

$$Q_1 = \frac{Q}{[(\text{FLC}_1/\text{FLC}_2)^{0.54} + 1]}$$

For the simplest of looped systems, i.e. a single loop, hand calculations are not complex. Furthermore, sometimes a seemingly complex piping system can be simplified by substituting an “equivalent pipe” for two or more pipes in series or in parallel.

For pipes in series

$$\text{FLC}_e = \text{FLC}_1 + \text{FLC}_2 + \text{FLC}_3 + \dots$$

For pipes in parallel

$$\left(\frac{1}{\text{FLC}_e}\right)^{0.54} = \left(\frac{1}{\text{FLC}_1}\right)^{0.54} + \left(\frac{1}{\text{FLC}_2}\right)^{0.54} + \dots$$

For gridded systems, which involve flow through multiple loops, computers are generally used since it becomes necessary to solve a system of nonlinear equations. When hand calculations are performed, the Hardy Cross [8] method of balancing heads is generally employed. This method involves assuming a flow distribution within the piping network, then iterating, i.e. applying successive corrective flows until differences in pressure losses through the various routes are nearly equal.

The Hardy Cross solution procedure applied to sprinkler system piping is as follows:

1. Identify all loop circuits and the significant parameters associated with each line of the loop, such as pipe length, diameter, and Hazen-Williams coefficient. Reduce the number of individual pipes where possible

by finding the equivalent pipe for pipes in series or parallel.

2. Evaluate each parameter in the proper units. Minor losses through fittings should be converted to equivalent pipe lengths. A value of all parameters except flow for each pipe section should be calculated (FLC).
3. Assume a reasonable distribution of flows that satisfies continuity, proceeding loop by loop.
4. Compute the pressure (or head) loss due to friction, p_f , in each pipe using the FLC in the Hazen-Williams formula.
5. Sum the friction losses around each loop with due regard to flow direction, i.e. assume clockwise flow positive and counter-clockwise flow as negative. Flows are correct when the sum of the losses, dp_f , is as small as desirable, typically 0.5 psi (0.03 bar).
6. If the sum of the losses is not sufficiently small for each loop, divide each pipe's friction loss by the presumed flow for the pipe, p_f/Q .
7. Calculate a correction flow for each loop as

$$dQ = \frac{-dp_f}{[1.85 \sum (p_f/Q)]}$$

8. Add the correction flow values to each pipe in the loop as required, thereby increasing or decreasing the earlier assumed flows. For cases where a single pipe is in two loops, the algebraic difference between the two values of dQ must be applied as the correction to the assumed flow.
9. With a new set of assumed flows, repeat steps 4–7 until the values of dp_f are sufficiently small.

10. As a final check, calculate the pressure loss by any route from the initial to the final junction. A second calculation along another route should give the same value of pressure loss within the range of accuracy expected, again typically 0.5 psi (0.03 bar).

Design standards typically require that pressures be shown to balance within 0.5 psi (0.03 bar) at hydraulic junction points. The designer, with or without the use of a computer program, must continue to make successive guesses as to how much flow takes place in each section of pipe until the pressure loss from the design area back to the source of supply is approximately the same (within 0.5 psi [0.03 bar]) regardless of the path chosen.

Example 10 For the small two-loop grid shown in Fig. 42.4, the total flow in and out is 100 gpm (378.5 L/min). It is necessary to determine the flow taking place through each pipe section. The system has already been simplified by finding the equivalent pipe for all pipes in series and in parallel.

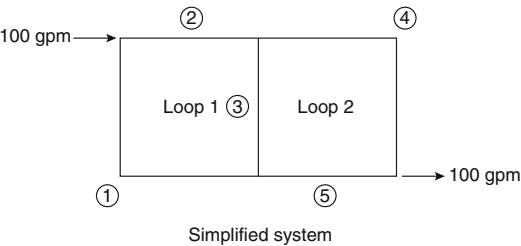


Fig. 42.4 Simplified system, pipe in series

- Pipe 1 FLC = 0.001
- Pipe 2 FLC = 0.002
- Pipe 3 FLC = 0.003
- Pipe 4 FLC = 0.001
- Pipe 5 FLC = 0.004

Recall that FLC pertains to the length, internal diameter, and Hazen-Williams C-factor for each pipe segment.

Under step 3 of the Hardy Cross procedure, flows that would satisfy conservation of mass are estimated as shown in Fig. 42.5. Steps 4–9 are then carried out in a tabular approach as shown in Table 42.3.

As the difference between dp_f for loop 1 and loop 2 is greater than 0.5 psi (0.03 bar), at least another iteration is necessary to balance flows. Using the calculated flow estimates from the first iteration, another set of calculations is completed as shown in Table 42.4. Revised flows after the first iteration are shown in Fig. 42.6.

With regard to the flow in pipe #3, which is common to both loops, the first iteration indicates that the estimated flow should be reversed. For loop 1, the flow calculates to –11.4 gpm but was estimated to be +5 gpm.

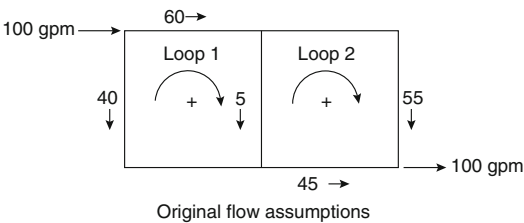


Fig. 42.5 Original flow assumptions

Table 42.3 First iteration

Loop	Pipe	Q	FLC	p_f	dp_f	(p_f/Q)	$dQ = -dp_f/1.85[\Sigma(p_f/Q)]$	$Q + dQ$
1	1	–40	0.001	–0.92		0.023	$dQ = -16.4$	–56.4
	2	60	0.002	3.90		0.065		+43.6
	3	5	0.003	0.06		0.012		–11.4
					=3.04	0.100		
2	3	–5	0.003	–0.06		0.012	$dQ = 11.2$	+6.2
	4	55	0.001	1.66		0.030		+66.2
	5	–45	0.004	–4.58		0.102		+33.8
					= –2.98	0.144		

Table 42.4 Second iteration

Loop	Pipe	Q	FLC	p_f	dp_f	(p_f/Q)	$dQ = -dp_f/1.85[\Sigma(p_f/Q)]$	$Q + dQ$
1	1	-56.4	0.001	-1.74		0.031		-54.0
	2	43.6	0.002	2.16		0.050		+46.0
	3	-22.6	0.003	-0.96		0.042	$dQ = 2.4$	-20.2
					$= -0.54$	0.123		
2	3	22.6	0.003	0.96		0.042		+20.5
	4	66.2	0.001	2.34		0.035		+64.1
	5	-33.8	0.004	-2.69		0.080		+35.9
					$= 0.61$	0.157	$dQ = -2.1$	

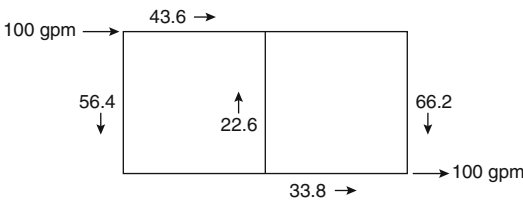


Fig. 42.6 Corrected flows after first iteration

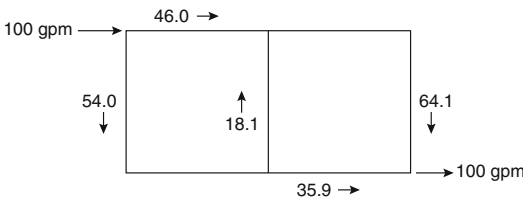


Fig. 42.7 Corrected flows after second iteration

For loop 2, the flow calculates +6.2 gpm but was estimated to be -5 gpm. As indicated in step 8, where a pipe segment is common to another loop, the algebraic difference between the two values of dQ is to be applied as the correction to the assumed flow. In other words, the flow correction for the common pipe is the net effect of the corrections for both loops. For pipe #3, the algebraic difference in dQ for loops 1 and 2 is 27.6 gpm (16.4 gpm + 11.2 gpm). This results in a corrected flow for pipe #3 of $5 - 27.6 = -22.6$ gpm for loop 1 and +22.6 for loop 2.

After completion of the second iteration, the difference in dp_f between loops 1 and 2 is still too large, so a third iteration is needed. The flows after the second iteration are shown in Fig. 42.7. The third iteration calculations, as shown in Table 42.5, indicate that the In starting the

pressure losses around both loops are balanced within 0.5 psi. Therefore, the flow split assumed after two iterations can be accepted. As a final check, step 10 of the above procedure calls for a calculation of the total pressure loss through two different routes, requiring that they balance within 0.5 psi (0.03 bar):

Water flow route through pipes 1 and 5:

$$\begin{aligned} & \text{FLC}_1 (Q_1)^{1.85} + \text{FLC}_2 (Q_2)^{1.85} \\ &= 0.001(54.0)^{1.85} + 0.004(35.9)^{1.85} \\ &= 1.6 + 3.0 = 4.6 \text{ psi (0.32 bar)} \end{aligned}$$

Water flow route through pipes 2 and 4:

$$\begin{aligned} & 0.002(46.0)^{1.85} + 0.001(64.1)^{1.85} = 2.4 + 2.2 \\ &= 4.6 \text{ psi (0.32 bar)} \end{aligned}$$

This is acceptable. Note that this example required less than three full iterations to achieve a successful solution, i.e. correctly balanced flow, despite the fact that the initial flow assumption called for reverse flow in pipe #3. The initial assumption was for a clockwise flow of 5 gpm (18.9 L/min) in pipe 3, but the final solution shows a counterclockwise flow of 18.1 gpm (68.5 L/min).

Water Supply Calculations

Determination of Available Supply Curve

Depending upon the location of the sprinkler system, public or private water main networks might be available to serve as the water supply

Table 42.5 Third iteration

Loop	Pipe	Q	FLC	p_f	dp_f	(p_f/Q)	$dQ = -dp_f/1.85[\Sigma(p_f/Q)]$	$Q + dQ$
1	1	-54.0	0.001	-1.60				
	2	46.0	0.002	2.38				
	3	-18.1	0.003	-0.64				
					=0.14			
2	3	18.1	0.003	0.64				
	4	64.1	0.001	2.20				
	5	-35.9	0.004	-3.01				
					= -0.17			

for the system. For instance, the municipal underground water mains of many large cities in North America are permitted to be used for fire protection purposes.

Flow testing of public or private water supply mains permits an evaluation of the strength of the available water supply in terms of both quantity of flow and available pressures. The strength of a water supply is the key to whether it will adequately serve a sprinkler system.

Each test of a water supply must provide at least two pieces of information—a static pressure and a residual pressure at a known flow. The static pressure is sometimes referred to as the “no flow” condition, as no water is being discharged from the main in the vicinity of the test. However, it must be recognized that rarely is any public water supply network in a true no flow condition. This condition is intended represent a situation where the fire protection system is not creating an additional flow demand beyond that which is ordinarily placed on the system. The residual pressure reading is taken with an additional flow being taken from the system, preferably a flow that approximates the likely maximum system demand.

Between the two (or more) points, a representation of the water supply (termed a water supply curve) can be made. For the most part, this water supply curve is a fingerprint of the system supply and piping arrangements, since the static pressure tends to represent the effect of elevated tanks and operating pumps in the system, and the drop to the residual pressure represents the friction and minor losses through the piping network that result from the increased flow during the test.

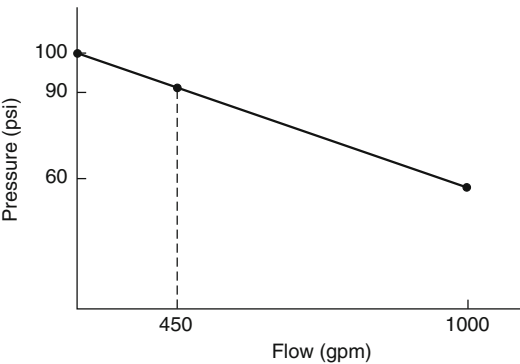


Fig. 42.8 Pressure available from 450 gpm flow water supply

The static pressure is read directly from a gauge attached to a hydrant. The residual pressure is read from the same gauge while a flow reading is taken from another hydrant, preferably downstream. A pitot tube is usually used in combination with observed characteristics of the nozzle through which flow is taken in order to determine the amount of flow.

Figure 42.8 is an example of a plot of water supply information. The static pressure is plotted along the y-axis, reflecting a given pressure under zero or no-flow conditions. The residual pressure at the measured flow is also plotted, and a straight line is drawn between these two points. Note that the x-axis is not linear, but rather shows flow as a function of the 1.85 power. This corresponds to the exponent for flow in the Hazen-Williams equation. Using this semi-exponential graph paper demonstrates that the residual pressure effect is the result of friction loss through the system, and permits the water supply curve to be plotted as a straight line.

Since the drop in residual pressure is proportional to flow to the 1.85 power, the available pressure at any flow can be read directly from the water supply curve.

For adequate design, the system demand point, including hose stream allowance, should lie below the water supply curve.

Example 11 If a water supply is determined by test to have a static pressure of 100 psi (6.9 bar) and a residual pressure of 80 psi (5.5 bar) at a flow of 1000 gpm (3785 L/min), the pressure available at a flow of 450 gpm (1703 L/min) can be approximated by plotting the two known data points on the hydraulic graph paper as shown in Fig. 42.8. At a flow of 450 gpm (1703 L/min), a pressure of 90 psi (6.2 bar) is indicated.

Pump Selection and Testing

Specific requirements for pumps used in sprinkler systems are normally contained in separate design and installation standards such as NFPA 20.

Fire pumps provide a means of making up for pressure deficiencies where an adequate volume of water is available at a suitable net positive suction pressure. Plumbing codes or municipal water supply regulations sometimes set a minimum allowable net positive suction pressure of 10–20 psi (0.69–1.38 bar) for water taken from public mains. If insufficient water is available at such pressures from such sources, then it becomes necessary to use a stored water supply.

Listed centrifugal fire pumps are available with either diesel or electric drivers, and with capacities ranging from 25 to 5000 gpm (95–18,927 L/min), although fire pumps are most commonly found with capacities ranging from 250 to 2500 gpm (946–9463 L/min) in increments of 250 up to 1500 gpm (946 up to 5678 L/min) and 500 gpm (1893 L/min) increments beyond that point. Each pump is specified with a rated flow and rated pressure. Rated pressures vary extensively, since manufacturers can control this feature with small changes to impeller design.

Pump affinity laws govern the relationship between impeller diameter, D , pump speed, N , flow, Q , pressure head, H , and brake horsepower, bhp. The first set of affinity laws assumes a constant impeller diameter.

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2} \quad \frac{H_1}{H_2} = \frac{N_1^2}{N_2^2} \quad \frac{\text{bhp}_1}{\text{bhp}_2} = \frac{N_1^3}{N_2^3}$$

These affinity laws are commonly used when correcting the output of a pump to its rated speed, such as during a fire pump acceptance test when the installed pump is not operating precisely at its rated speed.

The second set of the affinity laws assumes constant speed with change in impeller diameter, D .

$$\frac{Q_1}{Q_2} = \frac{D_1}{D_2} \quad \frac{H_1}{H_2} = \frac{D_1^2}{D_2^2} \quad \frac{\text{bhp}_1}{\text{bhp}_2} = \frac{D_1^3}{D_2^3}$$

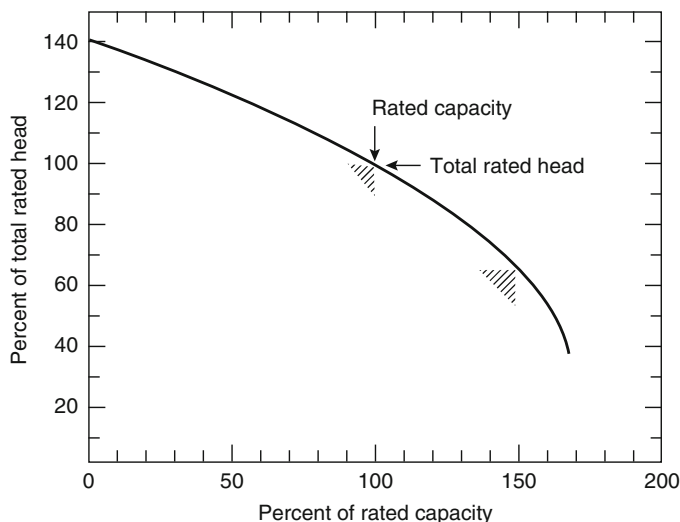
Pumps are selected to fit the system demands on the basis of three key points relative to their rated flow and rated pressure (Fig. 42.9). Fire pump standards such as NFPA 20 specify that centrifugal fire pumps meet these three points as noted below, and the listing laboratories verify this and establish pump performance curves for each pump.

1. A minimum of 100 % of rated pressure at 100 % of rated flow
2. A minimum of 65 % of rated pressure at 150 % of rated flow (overload)
3. A maximum of 140 % of rated pressure at 0 % of rated flow (churn)

While each fire pump has its individual performance curve, a pump specifier knows the basic performance characteristics of a pump even before the performance curve is available, since it must meet the three points described above. It is usually possible to have more than one option when choosing pumps, since the designer is not limited to using a specific point on the pump performance curve.

There are limits to flexibility in pump selection, however. For example, it is not permitted to install a pump in a situation where it would be expected to operate with a flow exceeding 150 % of rated capacity, since the performance is not a

Fig. 42.9 Pump performance curve



known factor, and indeed available pressure is usually quick to drop off beyond this point.

NFPA 20 has traditionally provided guidance on what part of the pump curve to use. A centrifugal fire pump should be selected in the range of operation from 90 % to 150 % of its rated capacity. The performance of the pump when applied at capacities over 140 % of rated capacity may be adversely affected by the suction conditions, but if suction conditions can be properly assured, the pump can operate at any point on its characteristic curve from shutoff to 150 % of its rated capacity.

Application of the pump at capacities less than 90 % of the rated capacity is not recommended.

Where specific pump performance curve is not available, the adequacy of a pump can be determined on the basis of the required performance points. For design capacities below the rated capacity, the rated pressure should be used. For design capacities between 100 % and 150 % of rated capacity, the pressure used should be found by the relationship made apparent by similar triangles.

$$\frac{0.35P}{0.5Q} = \frac{P' - 0.65P}{1.5Q - Q'}$$

where P and Q are the rated pressure and capacity, and P' is the minimum available pressure at capacity, Q' , where $Q < Q' < 1.5Q$.

Example 12 A pump is to be selected to meet a demand of 600 gpm (2271 L/min) at 85 psi (5.86 bar). To determine whether a pump rated for 500 gpm (1893 L/min) at 100 psi (6.90 bar) would be able to meet this point without having an actual pump performance curve to work from, the above formula can be applied, with $P = 100$, $Q = 500$, and $Q' = 600$.

Inserting these values gives

$$\begin{aligned} \frac{(0.35)(100)}{(0.5)(500)} &= \frac{[P' - (0.65)(100)]}{[(1.5)(500) - 600]} \\ \frac{35}{250} &= \frac{(P' - 65)}{(750 - 600)} \\ P' &= 65 + 21 = 86 \text{ psi (5.93 bar)} \end{aligned}$$

Since the value of P' so calculated is greater than the 85 psi (5.86 bar) required, the pump will be able to meet the demand point.

Tank Sizing

Tank selection and sizing are relatively straightforward compared to fire pump selection. The most basic question is whether to use a standalone elevated storage (gravity) tank, or a pressure tank, or a suction tank in combination with a fire pump. Standards such as NFPA 22 describe the types of tanks in terms of suitable

construction materials, and provide design and installation requirements.

From a calculation standpoint, tanks must be sized to provide the minimum durations specified by NFPA 13 or other applicable standards for the system design. Required pressures must still be available when the tanks are nearly depleted of their water supplies. Durations are based on consideration of the full hydraulic demand (i.e., all sprinklers flowing in the design area). This is a conservative assumption for an automatic sprinkler system, due to the fact that the design area itself is considered to include some conservatism, with the additional understanding that sprinkler operations take place incrementally. Because of this conservatism, it is not necessary that the duration also be provided for a hydraulically less demanding design area, which would be a design area closer to the tank. Minimum durations are generally based on hazard classification, with shorter minimum durations allowed for systems with remote alarm service to a constantly attended location.

If the tank is intended to provide the needed supply without the use of a fire pump, the energy, i.e. pressure for the sprinkler system must be available due to the height of the bottom of a gravity tank or the air pressure held within a pressure tank.

An important factor in gravity tank calculations is the requirement that the pressure available from elevation (calculated using 0.433 psi per foot [0.099 bar/m]) must be determined using the lowest expected level of water in the tank. This is normally the point at which the tank would be considered empty.

In sizing pressure tanks, the percentage of air in the tanks must be controlled so as to ensure that the last remaining quantity of water leaving the tank will be flowing at an adequate pressure. While a common rule of thumb has been that one-third of the tank's volume consist of air at a minimum pressure of 75 psi (5.17 bar), this rule does not hold true for systems with high pressure demands or where the tank is located a considerable distance below the level of the highest sprinkler.

For pipe schedule systems, two formulas have traditionally been used, based on whether the tank is located above the level of the highest sprinkler or some distance below.

For the tank above the highest sprinkler

$$P = \frac{30}{A} - 15$$

For the tank below the highest sprinkler

$$P = \left(\frac{30}{A} - 15 \right) + \left(\frac{0.434H}{A} \right)$$

where

A = Proportion of air in the tank

P = Air pressure carried in the tank in psi

H = Height of the highest sprinkler above the tank bottom in ft

It can be seen that these formulas are based simply on the need to provide a minimum pressure of 15 psi (1.03 bar) to the system at the level of the highest sprinkler, and an assumption of 15 psi (1.03 bar) atmospheric pressure.

Using the same approximation for atmospheric pressure, a more generalized formula has come into use for hydraulically designed systems:

$$P_i = \frac{P_f + 15}{A} - 15$$

where

P_i = Tank air pressure to be used

P_f = System pressure required per hydraulic calculations

A = Proportion of air in the tank

Example 13 A pressure tank is to be used to provide a 30 min water supply to a system with a hydraulically calculated demand of 140 gpm (530 L/min) at a pressure of 118 psi (8.14 bar). Due to nearby component pressure ratings, it is important that air pressure in the tank not exceed 175 psi (12.0 bar). To determine the minimum size tank that can be used, it is important not only to consider the total amount of water needed, but also the amount of air necessary to keep the pressures within the stated limits of 118 and 175 psi.

The above equation for hydraulically designed systems can be used to solve for A .

$$P_i = \left[\frac{(P_f + 15)}{A} \right] - 15$$

If

$$A = \frac{(P_f + 15)}{(P_i + 15)}$$

$$A = \frac{(118 + 15)}{(175 + 15)} = \frac{133}{190} = 0.70$$

then

This means that the tank will need to be 70 % air if the air pressure in the tank is to be kept to 175 psi (12.0 bar).

The minimum water supply required is 30 min $\times$ 140 gpm = 4200 gal (15,898 L).

Thus, the minimum tank volume will be such that 4200 gal (15,898 L) can be held in the remaining 30 % of volume.

$$0.3V = 4,200 \text{ gal}$$

$$V = \frac{4,200}{0.3} = 14,000 \text{ gal tank (53,000 L)}$$

Hanging and Bracing Methods

Hangers and Hanger Supports

Sprinkler design standards such as NFPA 13 contain a great deal of specific guidance relative to hanger spacing and sizing based on pipe sizes. It should be recognized that performance-based approaches are also often permitted. Different criteria can exist for individual hangers and their connection to the supporting building structure. For example, NFPA 13 considers any hanger and installation method is acceptable if certified by a registered professional engineer to meet the following criteria:

1. Hangers are capable of supporting five times the weight of the water-filled pipe plus 250 lb (114 kg) at each point of piping support.
2. Points of support are sufficient to support the sprinkler system.

3. Spacing between hangers does not exceed the limits within the standard for the various types of piping materials.
4. Ferrous materials are used for hanger components.
5. Detailed calculations are submitted when required by the reviewing authority.

The building structure itself must be capable of supporting the weight of the water-filled pipe plus 250 lbs (114 kg) applied at the point of hanging. The 250 lb (114 kg) weight is intended to represent the extra loading that would occur if a relatively heavy individual were to hang on the piping.

Trapeze Hangers

Trapeze hangers are used where structural members are not located, so as to provide direct support of sprinkler lines or mains. This can occur when sprinkler lines or mains run parallel to structural members such as joists or trusses.

Because they are considered part of the support structure, the criteria within NFPA 13 call for the hangers to support the weight of 15 ft (5 m) of water-filled pipe plus 250 lbs (114 kg) applied at the point of hanging. An allowable bending stress of 15 ksi (103 MPa) is used for steel members. Two tables are provided in the standard, one of which presents required section moduli based on the span of the trapeze and the size and type of pipe to be supported, and the other of which presents the available section moduli of standard pipes and angles typically used as trapeze hangers.

In using the tables, the standard allows the effective span of the trapeze hanger to be reduced if the load is not at the midpoint of the span. The equivalent length of trapeze is determined from the formula

$$L = \frac{4ab}{(a + b)}$$

where L is the equivalent length, a is the distance from one support to the load, and b is the distance from the other support to the load.

Example 14 A trapeze hanger is required for a main running parallel to two beams spaced 10 ft (3.048 m) apart. If the main is located 1 ft 6 in. (0.457 m) from one of the beams, the equivalent span of trapeze hanger required can be determined by using the formula

$$L = \frac{4(1.5 \text{ ft})(8.5 \text{ ft})}{(1.5 \text{ ft} + 8.5 \text{ ft})} = 5.1 \text{ ft}(1.554 \text{ m})$$

Earthquake Braces

Protection for sprinkler systems in earthquake areas is provided in several ways. Flexibility and clearances are added to the system where necessary to avoid the development of stresses that could rupture the piping. Too much flexibility could also be dangerous, however, since the momentum of the unrestrained piping during shaking could result in breakage of the piping under its own weight or on collision with other building components. Therefore, lateral and longitudinal bracing is required for all mains and lateral bracing is required for branch lines exceeding 2 in. (50 mm) in diameter. Smaller branch lines are required to be restrained against movement, which involves a less rigorous means of holding the piping in place.

Calculating loads for earthquake braces is based on the assumption that the normal hangers provided to the system are generally capable of handling vertical forces. However, the upward vertical component of strong horizontal forces must be addressed where braces are insufficiently angled from the horizontal. Traditionally, horizontal forces were conservatively approximated by a constant acceleration equal to one-half that of gravity.

$$a_h = 0.5g$$

Due to advances in earthquake engineering, more specific mapping of expected earthquake accelerations is now available. Current codes call for the design of mechanical building systems to be based on maximum short-period (0.2 s) accelerations expected for the 500 year earthquake. NFPA 13 (2013) contains a seismic

coefficient table that allows a simplified method by which these accelerations can be converted to the horizontal seismic load for braces using the formula

$$F_{pw} = C_p W_p$$

where F_{pw} is the force acting on the brace, C_p is the seismic coefficient selected in the table on the basis of short period response, and W_p is 1.15 times the weight of water-filled piping supported by the brace.

Table 42.6 contains some of the NFPA 13 seismic coefficients based on short period accelerations, where the horizontal accelerations S_s are expressed relative to gravity. The seismic coefficients are based on the assumption of fairly soft soil and other conservative assumptions.

Since the braces can be called on to act in both tension and compression, it is necessary not only to size the brace member to handle the expected force applied by the weight of the pipe in its zone of influence, but also to avoid a member that could fail as a long column under buckling.

The ability of the brace to resist buckling is determined through an application of Euler's formula. Tables provide loads based on maximum slenderness ratios of 100, 200, and 300. The 300 value corresponds to the maximum slenderness ratio generally used under steel construction codes for secondary framing members. This is expressed as

Table 42.6 Seismic coefficient table

S_s	C_p
0.33 or less	0.35
0.50	0.40
0.75	0.42
0.95	0.50
1.00	0.51
1.25	0.58
1.50	0.70
2.00	0.93
2.40	1.12
3.00	1.40

Source: NFPA 13, Table 9.3.5.9.3 (2013)

$$\frac{\ell}{r} \leq 300$$

where ℓ is the length of the brace and r is the least radius of gyration for the brace.

The least radius of gyration for some common shapes is as follows:

pipe

$$r = \frac{(r_o^2 + r_i^2)^{1/2}}{2}$$

rod

$$r = \frac{r}{2}$$

flat

$$r = 0.29h$$

Special care must be taken in the design of earthquake braces so that the load applied to any brace does not exceed the capability of the fasteners of that brace to the piping system or the building structure, and that the braces are attached only to structural members capable of supporting the expected loads.

Performance Calculations

Sprinkler Response as a Detector

Automatic sprinklers serve a dual function as both heat detectors and water distribution devices. As such, the response of sprinklers can be estimated using the same methods as for response of heat detectors (see Chap. 40). Care should be taken, however, that these calculations are used within their limitations. Factors pertaining to sprinkler orientation, air flow deflection, radiation effects, heat of fusion of solder links, and convection within glass bulbs are all considered to introduce minor errors into the calculation process. Heat conduction to the sprinkler frame and distance of the sensing mechanism below the ceiling have been demonstrated to be significant factors affecting response, but are ignored in some computer

models. Efforts have been made to quantify the prediction capability of the models, including DETACT-QS and the more recent Fire Dynamics Simulator (FDS) [9].

Modeling of sprinkler response can be useful, particularly when used on a comparative basis. Beginning with the 1991 edition, an exception within NFPA 13 permitted variations from the rules on clearance between sprinklers and ceilings "...provided the use of tests or calculations demonstrate comparable sensitivity and performance."

Example 15 Nonmetallic piping extending 15 in. (0.38 m) below the concrete ceiling of a 10-ft (3.048 m) high basement 100 ft by 100 ft (30.48 × 30.48 m) in size makes it difficult to place standard upright sprinklers within the 12 in. (0.30 m) required by NFPA 13 for unobstructed construction. Using the LAVENT [10] computer model, and assuming RTI values of $400 \text{ ft}^{1/2} \times \text{s}^{1/2}$ ($221 \text{ m}^{1/2} \times \text{s}^{1/2}$) for standard sprinklers and $100 \text{ ft}^{1/2} \times \text{s}^{1/2}$ ($55 \text{ m}^{1/2} \times \text{s}^{1/2}$) for quick-response sprinklers, it can be demonstrated that the comparable level of sensitivity can be maintained at a distance of 18 in. (0.457 m) below the ceiling. Temperature rating is assumed to be 165 °F, and maximum lateral distance to a sprinkler is 8.2 ft (2.50 m) (10 ft × 13 ft [3.048 m × 3.962 m] spacing). Assuming the default fire (empty wood pallets stored 5 ft [1.52 m] high), for example, the time of actuation for the standard sprinkler is calculated to be 200 s, as compared to 172 s for the quick-response sprinkler. Since the noncombustible construction minimizes concern relative to the fire control performance for the structure, the sprinklers can be located below the piping obstructions.

Dry System Water Delivery Time

Total water delivery time consists of two parts. The first part is the trip time taken for the system air pressure to bleed down to the point where the system dry valve opens to admit water to the piping. The second part is the transit time for

the water to flow through the piping from the dry valve to the open sprinkler. In other words

Water delivery time = Trip time + Transit time

where water delivery time commences with the opening of the first sprinkler.

Sprinkler standards such as NFPA 13 have traditionally not contained a maximum water delivery time requirement if system volume is limited, generally to no more than 750 gal (2839 L). Larger systems were permitted only if water flow from a remote inspector's test connection took place within 60 s. As such, the rule of thumb for dry system operation is that no more than a 60-s water delivery time should be tolerated, and that systems should be divided into smaller systems if necessary to achieve this 1-min response. Beginning with the 2007 edition of NFPA 13, the 60-s water delivery time was mandated for dry pipe systems protecting dwelling unit portions of any building. Dry system response is simulated in field testing by the opening of an inspector's test connection. The inspector's test connection is required to be at the most remote point of the system from the dry valve, and is required to have an orifice opening of a size simulating the smallest orifice sprinkler installed on the system.

The water delivery time of the system is recorded as part of the dry pipe valve trip test that is conducted using the inspector's test connection. However, it is not a realistic indication of actual water delivery time for two reasons:

1. The first sprinkler to open on the system is likely to be closer to the system dry valve, reducing water transit time.
2. If additional sprinklers open, the trip time will be reduced since additional orifices are able to expel air. Water transit time may also be reduced since it is easier to expel the air ahead of the incoming water.

FM Global researchers have shown [11] that it is possible to calculate system trip time using the relation

$$t = 0.0352 \frac{V_T}{A_n T_0^{1/2}} \ln \left(\frac{p_{a0}}{p_a} \right)$$

where

t = Time (s)

V_T = Dry volume of sprinkler system (ft³)

T_0 = Air temperature (°R)

A_n = Flow area of open sprinklers (ft²)

p_{a0} = Initial air pressure (absolute)

p_a = Trip pressure (absolute)

Calculating water transit time is more difficult, but may be accomplished using mathematical models. FM Global researchers developed the first such model in the 1970s. In 2003 a dry system water delivery model was introduced to the commercial marketplace [12], following the incorporation of acceptance criteria into the 2002 edition of NFPA 13. A literature review conducted in 2007, intended to develop data to assist in the evaluation of the traditional 60-s water delivery requirement of NFPA 13, observed that “the water delivery time limit of 60 s has some, but not overwhelming data to support requiring or not requiring a time limitation for small systems” [13].

Droplet Size, Penetration and Motion

For geometrically similar sprinklers, the median droplet diameter in the sprinkler spray has been found to be inversely proportional to the $^{1/3}$ power of water pressure and directly proportional to the $^{2/3}$ power of sprinkler orifice diameter such that

$$d_m \propto \frac{D^{2/3}}{p^{1/3}} \propto \frac{D^2}{Q^{2/3}}$$

where

d_m = Mean droplet diameter

D = Orifice diameter

P = Pressure

Q = Rate of water flow

The relationship of droplet size production to pressure and orifice diameter has been confirmed using high-magnification shadow imaging [14].

A sprinkler “penetration ratio” has likewise been observed to be proportional to the median droplet diameter, which is needed for fire plume penetration when the sprinkler spray is in the

gravity mode [15]. However, fire plumes can also be penetrated by water sprays using a momentum mode. Fire plume penetration is considered essential to the early suppression of the fire by automatic sprinklers, as compared to fire control, in which the spread of the fire is stopped and the impact mitigated.

Total droplet surface area has been found to be proportional to the total water discharge rate divided by the median droplet diameter

$$A_s \propto \frac{Q}{d_m}$$

where A_s is the total droplet surface area.

Combining these relationships, it can be seen that

$$A_s \propto (Q^3 p D^{-2})^{1/3}$$

When a droplet with an initial velocity vector of U is driven into a rising fire plume, the one-dimensional representation of its motion has been represented as [16]

$$\frac{m_1 dU}{dt} = m_1 g - \frac{C_D \rho_g (U + V)^2}{2S_f}$$

where

U = Velocity of the water droplet

V = Velocity of the fire plume

m = Mass of the droplet

ρ_g = Density of the gas

g = Acceleration of gravity

C_D = Coefficient of drag

S_f = Frontal surface area of the droplet

The first term on the right side of the equation represents the force of gravity, while the second term represents the force of drag caused by gas resistance. The drag coefficient for particle motion has been found empirically to be a function of the Reynolds number (Re) as [17]

$$C_D = 18.5 \text{ Re}^{-0.6} \text{ for } \text{Re} < 600$$

$$C_D = 0.44 \text{ for } \text{Re} > 600$$

Mathematical modeling comparing the drag force of a sprinkler spray to the buoyancy of a smoke layer in the vicinity of sprinklers, validated by full scale experiments, has been

used recently to address the long-standing question of the value of automatic smoke vents in sprinklered buildings. The modeling indicated that increases in sprinkler operating pressure eventually lead to ineffective smoke venting, and that the area of smoke venting has very little influence on smoke flow once sprinkler operation causes a loss in smoke flow efficiency [18].

Spray Density and Cooling

The heat absorption rate of a sprinkler spray is expected to depend on the total surface area of the water droplets, A_s , and the temperature of the ceiling gas layer in excess of the droplet temperature, T . With water temperature close to ambient temperature, T can be considered excess gas temperature above ambient.

Chow [19] has developed a model for estimating the evaporation heat loss due to a sprinkler water spray in a smoke layer. Sample calculations indicate that evaporation heat loss is only significant for droplet diameters less than 0.5 mm. For the droplet velocities and smoke layer depths analyzed, it was found that the heat loss to evaporation would be small (10–25 %), compared to the heat loss from convective cooling of the droplets.

FM Global researchers [20] have developed empirical correlations for the heat absorption rate of sprinkler spray in room fires, as well as convective heat loss through the room opening, such that

$$\dot{Q} = \dot{Q}_{\text{cool}} + \dot{Q}_c + \dot{Q}_l$$

where

$\dot{Q}$ = Total heat release rate of the fire

$\dot{Q}_{\text{cool}}$ = Heat absorption rate of the sprinkler spray

$\dot{Q}_c$ = Convective heat loss rate through the room opening

$\dot{Q}_l$ = Sum of the heat loss rate to the walls and ceiling, $\dot{Q}_s$, the heat loss rate to the floor, $\dot{Q}_f$, and the radiative heat loss rate through the opening, $\dot{Q}_r$

Test data indicated that

$$\dot{Q}_{\text{cool}}/\dot{Q} = 0.000039\Lambda^3 - 0.003\Lambda^2 + 0.082\Lambda$$

For

$$0 < \Lambda \leq 33 / (\min \times \text{kW}^{1/2} \times \text{m}^{5/4})$$

where Λ is a correlation factor incorporating heat losses to the room boundaries and through openings as well as to account for water droplet surface area.

$$\Lambda = (AH^{1/2}\dot{Q}_l)^{-1/2} (W^3PD^{-2})^{1/3}$$

for

$$P = \frac{p}{(17.2 \text{ kPa})} \quad \text{and} \quad D = \frac{d}{0.0111 \text{ m}}$$

where

A = Area of the room opening (m)

H = Height of the room opening (m)

P = Water pressure at the sprinkler (bar)

d = Sprinkler nozzle diameter (m)

W = Water discharge (L/min)

The above correlations apply to room geometry with length-to-width ratio of 1.2–2 and opening size of 1.70–2.97 m².

Suppression by Sprinkler Sprays

In 1993, researchers at the National Institute of Standards and Technology (NIST) developed a “zeroth order” model of the effectiveness of sprinklers in reducing the heat release rate of furnishing fires [21]. Based on measurements of wood crib fire suppression with pendant spray sprinklers, the model was described as conservative. The model assumed that all fuels have the same degree of resistance to suppression as a wood crib, despite the fact that tests have shown furnishings with large burning surface areas can be extinguished easily compared to the deep-seated fires encountered with wood cribs.

The recommended equation, which relates to fire suppression for a 610-mm-high crib, has also been checked for validity with 305 mm crib results. The equation is

$$\dot{Q}(t - t_{\text{act}}) = \dot{Q}(t_{\text{act}}) \exp \left[\frac{-(t - t_{\text{act}})}{3.0(\dot{w}'')^{-1.85}} \right]$$

where

$\dot{Q}$ = Heat release rate (kW)

t = Any time following t_{act} of the sprinklers (s)

$\dot{w}''$ = Spray density (mm/s)

The NIST researchers claimed the equation was appropriate for use where the fuel is not shielded from the water spray, and the application density is at least 0.07 mm/s (4.2 mm/min [0.1 gpm/ft²]). The method does not account for variations in spray densities or suppression capabilities of individual sprinklers.

The model must be used with caution, since it was developed on the basis of fully involved cribs. It does not consider the possibility that the fire could continue to grow in intensity following initial sprinkler discharge, and, for that reason, should be restricted to use in light hazard applications.

Sprinklers are assumed to operate within a room of a light hazard occupancy when the total heat release rate of the fire is 500 kW. The significance of an initial application rate of 0.3 gpm/ft² (0.205 mm/s) as compared to the minimum design density of 0.1 gpm/ft² (0.07 mm/s) can be evaluated by the expected fire size after 30 s. With the minimum density of 0.07 mm/s (0.1 gpm/ft²), the fire size is conservatively estimated as 465 kW after 30 s. With the higher density of 0.205 mm/s (0.3 gpm/ft²), the fire size is expected to be reduced to 293 kW after 30 s. Corresponding values after 60 s are 432 and 172 kW, respectively.

More recent efforts to model suppression by automatic sprinklers have taken place as part of the NIST's development of the Fire Dynamics Simulator (FDS) computational fluid dynamics model. Within that model, simulating the effects of a sprinkler spray involves predicting activation, computing droplet trajectories, and tracking water as it drips onto the burning fuel. In order to compute droplet trajectories, the initial size and velocity of each droplet must be estimated, a process that is one of the limiting factors in the use of the model for practical applications. As stated earlier, efforts have been under way to

develop atomization models for integration with FDS to better characterize the formation and distribution of droplets from the impact of a water stream on a sprinkler deflector [22, 23]. The effect of droplets on burning surfaces is another area in which additional work is needed. When dealing with liquid droplets hitting a solid surface, the current Version 5 of FDS assigns a random horizontal direction and moves at a fixed velocity on the order of 0.5 m/s until it reaches the edge, at which point it is assumed to drop down at the same velocity. While on the surface, the droplet is assumed to contribute to the formation of a surface film of water that participates in heat transfer. If the surface is burning, assumptions need to be made about the extent to which the water reduces the pyrolysis rate of the fuel. Most of the available correlations are based on fires involving rack storage of standard commodities in corrugated cartons. Work at Factory Mutual [24] has led to the following expression:

$$\dot{Q} = \dot{Q}_0 e^{k(t-t_0)}$$

where $\dot{Q}_0$ is the total heat release at the time of water application t_0 and k is a fuel-dependent constant, which in turn is dependent on the rate of water application.

Nomenclature

C	coefficient of friction
FLC	friction loss coefficient
Q	flow (gpm)
$\dot{Q}$	heat release rate (kW)
$\dot{W}''$	spray density (mm/s)
C_p	seismic coefficient
K	fuel-dependent constant

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